CHROMATIC HOMOTOPY THEORY SEMINAR MÜNSTER

Disclaimer: You can follow these notes during the talk or extend them or prepare your own. Essential references and content to be covered are always written at the beginning of each talk. All gray text is optional and can be easily skipped as it only provides some additional or background material which simplifies the understanding (in the author's opinion).

1. Talk 2. Equivalences of Formal Groups

- Briefly outline the picture from differential geometry: the beginning of Lecture 2 [1] and I.'s notes or any other source.
- Introduce étale sheaves as geometric setting, follow Lecture 2 [1], skip the part about the category of elements: [Definition 2.1], [Remark 2.2], some examples, [Definition 2.16] and below.
- Define formal groups and show how formal group laws produce formal groups, follow Lecture 3 [1]: [Definition 3.1], [Proposition 3.5], only if time permits (it is not presented in the notes, for curious [1]), [Definition 3.7], [Lemma 3.9], [Definition 3.14]
- Work out the example of complex-oriented cohomology theory and KU if time permits: [Example 3.17] and [Remark 3.18] [1], respectively. Show how $\mathbb{G}_m$ and $\mathbb{G}_a$ arise as formal groups associated to KU and $H\mathbb{Z}$ respectively.
- Sketch that over $\mathbb{Q}$ all formal group laws are equivalent: [Theorem 5.13] [1] using an intuition from differential geometry and taking some statements on faith (such as [Lemma 5.2] and [Lemma 5.11]).

In the previous talk, we defined a *formal group law* over a ring R to be a power series $F(x,y) \in R[[x,y]]$ which is unital, associative and commutative in the sense that

$$\begin{aligned} &(1) \quad F(x,y) = F(y,x), \\ &(2) \quad F(0,x) = F(x,0) = x, \\ &(3) \quad F(F(x,y),z) = F(x,F(y,z)). \end{aligned}$$

Observe that the axioms satisfied by F are reminiscent of it being a Taylor expansion of multiplication in a neighbourhood of the identity of some Lie group; that is, as if F defined a multiplication on some geometric object. See Appendix 2 for more details.

Our goal in this lecture will be to set up the language necessary to make this heuristic precise.

One geometric extension of commutative rings is given by the theory of schemes; unfortunately, this will not quite be sufficient for our purposes. Roughly, the issue is that F is a power series, rather than a polynomial, and so for it to define a multiplication we need some notion of convergence.

Compare with the end of the Appendix.

Multiplications on the 1-dimensional affine line over a ring R:

$$\mathbb{A}^1 \times_R \mathbb{A}^1 \longrightarrow \mathbb{A}^1$$

are given by choosing a polynomial:

$$R[t] \longrightarrow R[x] \otimes R[y] \cong R[x,y]$$

rather than a power series. Analogously for higher-dimensional affine spaces.

Date: WS 26–27.

An insight of Grothendieck is that more complicated geometric structures can be described by specifying how they interact with the simple ones, such as affine schemes; this is the functor-of-points approach. In concrete terms, this means that if X is our geometric object, perhaps something more general than a scheme, then the functor on affine schemes

$$y(X)(\mathrm{Spec}(A)) := \mathrm{Hom}_{\mathcal{G}}(\mathrm{Spec}(A), X)$$

where $\mathcal{G}$ is a category of geometric objects containing all affine schemes, should retain complete information about X. This means that our study can be rephrased in terms of such functors, and our more general class of "geometric objects" can be defined as a subclass of functors of this type.

Definition 1.1. An *étale sheaf* is a sheaf with respect to the étale topology on affine schemes:

$$X\colon (\mathsf{Aff})^{op} \longrightarrow \mathsf{Set}$$

The concrete description is given by the following:

Definition 1.2. An *étale sheaf* is a functor

$$X\colon \mathsf{CRing} \longrightarrow \mathsf{Set}$$

which satisfies *étale descent*; that is, for any finite set of étale maps (intuitively this means maps that are local diffeomorphisms, however in AG this is implemented in more technical terms, see [2]) $A \to B_i$ of commutative rings such that $A \to \prod_i B_i$ is faithfully flat [3] (intuitively this means that the collection of $\mathrm{Spec}(B_i)$ covers $\mathrm{Spec}(A)$), the diagram

$$X(A) \longrightarrow \prod_i X(B_i) \rightrightarrows \prod_{i,j} X(B_i \otimes_A B_j)$$

is an equalizer.

Background: Grothendieck topology is not quite a topology in the usual sense, it is data mimicking the structure of coverings, precisely:

Definition 1.3. A *Grothendieck topology* on a category $\mathcal{C}$ is a collection of covering families, satisfying the following axioms:

(1) If $f : Y \to X$ is an isomorphism, then $\{f\}$ is a covering family of X.
(2) If $\{f_i : U_i \to X\}$ is a covering family of X and $g : Y \to X$ is any morphism, then the fiber products $U_i \times_X Y$ exist and the projections $\{U_i \times_X Y \to Y\}$ form a covering family of Y.
(3) If $\{f_i : U_i \to X\}$ is a covering family of X and for each i, $\{g_{ij} : V_{ij} \to U_i\}$ is a covering family of U_i, then the compositions $\{f_i \circ g_{ij} : V_{ij} \to X\}$ form a covering family of X.

Example 1.4. The étale topology on the category of affine schemes is defined by declaring a family of morphisms $\{f_i : \mathrm{Spec}(B_i) \to \mathrm{Spec}(A)\}$ to be a covering family if each f_i is étale and the induced map $\mathrm{Spec}(\prod_i B_i) \to \mathrm{Spec}(A)$ is surjective.

Remark 1.5. For the purpose of our course the choice of the étale topology is not terribly important, although it is the standard one in some sources. Most of the properties we will be interested in can be verified in the (weaker) Zariski topology, and most of the sheaves we consider satisfy descent with respect to the (stronger) flat topology.

Personally I don't have good intuition for different (Grothendieck) topologies on schemes, so I tend to think of étale topology as the notion familiar from sheaves on topological spaces. This might be a wrong way to think about it, so I would be happy to hear some explanations or (if there are volunteers) add contributions for these notes.

Since it is sheaves of the type above that will be our language of choice, it is helpful to change our terminology to reflect this change of viewpoint.

Example 1.6. If B is a ring, then the corresponding *affine scheme* is the sheaf $\mathrm{Spec}(B)$

$$\mathrm{Spec}(B)(A) := \mathrm{Hom}_{\mathsf{CRing}}(B, A),$$

the set of homomorphisms of commutative rings.

Example 1.7. There is a fully faithful embedding of the category of schemes into the category of étale sheaves, given by:

$$y(S)(A) \coloneqq \mathrm{Hom}_{LocRingSp}(\mathrm{Spec}_{Zar}(A), S)$$

In particular we have:

Example 1.8. The *affine line* $\mathbb{A}^1$ is the sheaf defined by

$$\mathbb{A}^1(R) = R \cong \mathrm{Hom}_{\mathsf{CRing}}(\mathbb{Z}[t], R).$$

Example 1.9. The *multiplicative group* $\mathbb{G}_m$ is the sheaf defined by

$$\mathbb{G}_m(R) = R^\times \cong \mathrm{Hom}_{\mathsf{CRing}}(\mathbb{Z}[t^{\pm 1}], R),$$

It has a canonical structure of an abelian sheaf; that is, it lifts to a functor into abelian groups, corresponding to multiplication of units.

Remark 1.10. Definition 1.2 makes sense for functors $X\colon \mathrm{CRing} \to \mathcal{C}$ with values in any ∞-category $\mathcal{C}$ with limits, if the equalizer is replaced by the limit over the whole Čech nerve $X(B) \rightrightarrows X(B \otimes_A B) \cdots$.

For $\mathcal{C} = \mathrm{An}$ one obtains what is called (higher) stacks;

for $\mathcal{C} = \mathrm{Sp}$ one obtains sheaves of spectra, which are used, in particular, in the construction of topological modular forms: TMF, and this story goes very deep [4].

Now we are ready to define objects which will serve as open neighbourhoods on which formal group laws are multiplications.

Definition 1.11. Let A be a ring and let I be an ideal, which we can identify with an affine scheme $X = \mathrm{Spec}(A)$ together with a choice of a closed subscheme $Z = \mathrm{Spec}(A/I)$. Then, the *formal completion* of X along Z is the sheaf $\widehat{X}_Z$ defined by

$$\widehat{X}_Z(B) = \mathrm{colim}_n \mathrm{Hom}_{\mathsf{CRing}}(A/I^n, B).$$

A fancier way to state the previous definition is:

$$\widehat{X}_Z \cong \text{"}\varinjlim_n\text{"}\, \mathrm{Spec}(A/I^n) \in \mathrm{Ind}(Sch)$$

with arrows given by the natural maps $y_{\mathrm{Spec}(A/I^n)} \to y_{\mathrm{Spec}(A/I^{n+1})}$.

That is, $\widehat{X}_Z$ is the subfunctor of X consisting of those ring homomorphisms $A \to B$ which annihilate a power of I. An *affine formal scheme* is a sheaf isomorphic to one which arises in this process.

Note that we can equip the ring A with a linear topology where the basis of neighbourhoods of zero is given by the powers I^n. This makes A into a topological ring, and one can check that we have an identification

$$\mathrm{colim}_n \mathrm{Hom}_{\mathsf{CRing}}(A/I^n, B) \cong \mathrm{Hom}^{\mathrm{cont}}_{\mathsf{CRing}}(A, B),$$

where the right-hand side is given by *continuous* ring homomorphisms, where B is considered as having discrete topology.

Indeed, the basis of a topology on A is given by sets of the form $\{a+I^n : a \in A, n \geq 1\}$. Therefore, for a continuous map $f : A \to B$, the preimage of $0 \in B$ must be an open neighbourhood of $0 \in A$, which means it contains an ideal of the form I^n for some n.

And the fact that this topology is given by translations of the basis of neighbourhoods of 0 implies the other direction of the isomorphism.

Remark 1.12. Observe that the right-hand side given above, consisting of continuous ring homomorphisms, only depends on A and its topology, and so this formal completion is also called the *formal spectrum* of the topological ring A and denoted by $\mathrm{Spf}(A)$.

Remark 1.13. One useful perspective on the formal completion $\widehat{X}_Z$ is that it can be thought of as the *infinitesimal neighbourhood* of Z in X. Namely, suppose we have a map of rings $A \to B$ which factors through one of the A/I^n. Then, at the level of prime spectra, the continuous map

$$\mathrm{Spec}_{\mathrm{Zar}}(B) \longrightarrow \mathrm{Spec}_{\mathrm{Zar}}(A)$$

of topological spaces factors through the closed subspace $\mathrm{Spec}_{\mathrm{Zar}}(A/I)$. It does not usually factor in this way as a map of schemes, but the above suggests that it is "infinitesimally close" to one that does.

If you don't have any intuition of what is meant by "infinitesimally close" in the previous remark, I suggest to read the chapter 4.2 *Visualizing schemes II: nilpotents* [5].

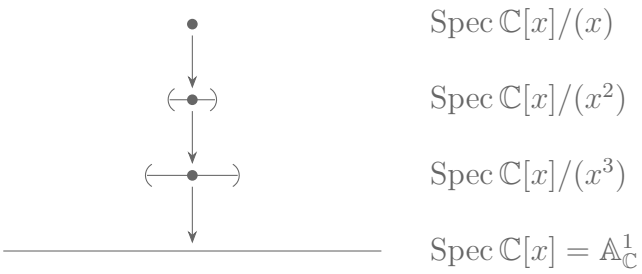


FIGURE 1. Picturing quotients of $\mathbb{C}[x]$ (see R. Vakil, *The Rising Sea* [5], Figure 4.2).

Remark 1.14. Two different ideals $I, J \subset A$ may very well give rise to the same topology (for example, one can take $J = I^2$) and it is important to remember that $\mathrm{Spf}(A)$ only depends on the topology and not on the choice of an ideal.

Because of the above remark, we can alternatively think of affine formal schemes as geometric objects associated to a (certain class of) topological rings.

Example 1.15. If R is a ring, then the formal affine line $\widehat{\mathbb{A}}^1_R$ over R is given by the following:

$$\widehat{\mathbb{A}}^1_R(A) = \mathrm{Hom}^{cont}_{CRing_R}(R[[t]], A),$$

where $R[[t]]$ is equipped with the t-adic topology, that is, the topology generated by the ideal (t).

Notice that there is another description of the formal affine line:

$$\mathrm{Hom}^{cont}_{CRing_R}(R[[t]], A) = \varinjlim_n \mathrm{Hom}_{CRing_R}(R[[t]]/(t^n), A) = \mathrm{Nil}(A),$$

where $\mathrm{Nil}(A)$ is the set of nilpotent elements of A.

This partially sheds light on the importance of nilpotent elements in algebraic geometry as elements which are "infinitesimally close" to zero.

Finally, we see how these new geometric objects help us make precise the intuition that a formal group law is a multiplication.

Definition 1.16. Let $F \in R[[x,y]]$ be a formal group law. Then, the corresponding *formal group* $\mathbf{G}_F$ is the abelian group object

$$\mathbf{G}_F \colon \mathrm{CRing}_R \longrightarrow \mathrm{Ab}$$

in étale sheaves over $\mathrm{Spec}(R)$ defined by

$$\mathbf{G}_F(B) = \mathrm{Nil}(B),$$

the set of nilpotent elements of B, with multiplication given by

$$(b_1, b_2) \longmapsto F(b_1, b_2).$$

Clearly, there are no issues with convergence here, because when b_1, b_2 are nilpotent, then $F(b_1, b_2)$ involves only finitely many non-zero terms.

The important thing to notice is that as a sheaf, $\mathbf{G}_F \cong \widehat{\mathbb{A}}^1_R$ - the formal affine line over R. Thus, analogously to the case of a Lie group (Appendix 2), we see that $\mathbf{G}_F$ is a (1-dimensional) formal neighbourhood on which F is forced to be a multiplication.

In fact, any abelian group structure on the formal affine line $\widehat{\mathbb{A}}^1_R$ over R with unit 0 comes from a unique formal group law over R.

Notation. *To work with formal group laws, it is convenient to introduce the notation*

$$x +_F y := F(x,y).$$

The axioms of being a formal group law can then be conveniently restated as

(1) $x +_F y = y +_F x$*;*
(2) $x +_F 0 = 0 +_F x = x$*;*
(3) $(x +_F y) +_F z = x +_F (y +_F z)$*.*

Despite its simplicity, the second axiom is very important, because it implies that

$$x +_F y = x + y + \text{higher order terms.}$$

Analogously to the case of a Lie group (Appendix 2).

While the definition of a formal group $\mathbf{G}_F$ states that it is an abelian group object, this is not transparent from the axioms of a formal group law alone and needs to be verified.

Lemma 1.17. *For any formal group law $F \in R[[x,y]]$, for any R-algebra B, the multiplication $(b_1, b_2) \mapsto b_1 +_F b_2$ makes* $\mathrm{Nil}(B)$ *into an abelian group.*

Proof. We have to show that any nilpotent $b \in B$ has an inverse. Let us prove this by induction on the lowest n such that $b^n = 0$, the case of $n = 1$ being easy since 0 is the unit and so has an inverse. If $n > 1$, then by expanding out, we see that

$$b +_F (-b) = b - b + b^2 x = b^2 x$$

for some element $x \in B$. However, $(b^2)^{n-1} = 0$, so that the latter has an inverse by the inductive assumption. This ends the argument. □

We are now ready to define one of the central objects of this seminar.

Definition 1.18. Let R be a ring. Then, a *formal group G over* $\mathrm{Spec}(R)$ is an abelian group object in étale sheaves over $\mathrm{Spec}(R)$ which is locally in the Zariski topology on R of the form $\mathbf{G}_F$, where F is a formal group law.

That is, a formal group is an abelian group object which "locally" comes from a formal group law; one can be very explicit about what this exactly means. Namely, an étale sheaf

$$G\colon \mathsf{CRing}_R \longrightarrow \mathsf{Ab}$$

is a formal group if there exists a finite list of elements $f_i \in R$ which jointly generate the unit ideal such that for any i, the restriction of the above functor to R_{f_i}-algebras is isomorphic to $\mathbf{G}_{F_i}$, where F_i is a formal group law over R_{f_i}.

We will see later that while there do exist formal groups that do not come from a formal group law, it is easy to tell whether one does, and in many situations it is enough to focus on the ones that do. For example, when R is local, then any formal group comes from a formal group law, because the only way f_i can generate a unit ideal is for one of them to be a unit.

The important part of the definition of a formal group is rather that it is "coordinate-free": even when there exists an isomorphism $G \cong \mathbf{G}_F$, it is *not* fixed. This leads to more natural statements.

Example 1.19. The *additive formal group* $\hat{\mathbb{G}}_a$ is defined by

$$\hat{\mathbb{G}}_a(B) := \mathrm{Nil}(B),$$

with the group structure given by ordinary addition. This is isomorphic to the formal group corresponding to the formal group law $F(x, y) = x + y$.

Example 1.20. The *multiplicative formal group* $\hat{\mathbb{G}}_m$ is defined by

$$\hat{\mathbb{G}}_m(B) := \{1 + b \mid b \text{ is nilpotent}\},$$

with multiplication induced by multiplication in the ring B. This is isomorphic to the formal group associated to the *multiplicative formal group law* $F(x, y) = x + y + xy$.

The above two examples are somewhat simple, because in both cases it is very easy to write down an isomorphism with a formal group corresponding to a simple formal group law.

The following two examples are more interesting because they are common situations where there is no canonical or even natural coordinate to choose.

Example 1.21. Let E be a complex-orientable cohomology theory, and for simplicity let's assume that the coefficient ring E^* is concentrated in even degrees, so that the underlying ungraded ring is commutative (since no Koszul signs arise). Then, $\mathrm{Spf}(E^*(\mathbb{CP}^\infty))$ is a formal group over $\mathrm{Spec}(E^*)$.

Here, we treat $E^*(\mathbb{CP}^\infty)$ as a topological ring with respect to the limit topology induced by the isomorphism

$$E^*(\mathbb{CP}^\infty) \cong \varprojlim_n E^*(\mathbb{CP}^n),$$

and the multiplication is the one induced by the map

$$\mathbb{CP}^\infty \times \mathbb{CP}^\infty \longrightarrow \mathbb{CP}^\infty$$

classifying the tensor product of line bundles.

Indeed, since E is complex-orientable, we've seen in the first lecture that a choice of a complex orientation $t \in E^2(\mathbb{CP}^\infty)$ will determine an isomorphism

$$E^*(\mathbb{CP}^\infty) \cong E^*[[t]].$$

This isomorphism is compatible with topologies on both sides and so lifts to one of formal schemes under which the multiplication on $\mathrm{Spf}(E^*(\mathbb{CP}^\infty))$ will correspond to the one coming from the formal group law which we denoted by F_E.

Note that the formal group law F_E depended on a choice of a complex-orientation, but the formal group $\mathrm{Spf}(E^*(\mathbb{CP}^\infty))$ does not. It is an *intrinsic invariant* of the complex-orientable cohomology theory E.

Remark 1.22. One can show that the formal groups corresponding to the two examples we considered, integral cohomology $H\mathbb{Z}$ and complex K-theory KU, are isomorphic to, respectively, the additive and multiplicative group. However, we will later see that there are many complex-oriented cohomology theories for which writing down the corresponding formal group law explicitly is impractical.

Again in the spirit of the analogy with Lie groups, we can define (announced) homomorphisms of formal groups:

Definition 1.23. Let $F, G \in R[[x, y]]$ be formal group laws. A *homomorphism* $f\colon F \to G$ is a power series $f(x) \in R[[x]]$ with $f(0) = 0$ and

$$f(x +_F y) = f(x) +_G f(y).$$

It is an *isomorphism* if it is invertible under composition, equivalently $f'(0) \in R^\times$, and a *strict isomorphism* if moreover $f'(0) = 1$.

Remark 1.24. Homomorphisms $F \to G$ are in bijection with homomorphisms $\mathbf{G}_F \to \mathbf{G}_G$ of formal groups over $\mathrm{Spec}(R)$. In particular, changing the coordinate on $\mathbf{G}_F$ by an invertible f replaces F by the isomorphic formal group law $f(F(f^{-1}(x), f^{-1}(y)))$; this is the precise meaning of "coordinate-free" above. Compare to Appendix 2

Example 1.25. As was shown for $E = H\mathbb{Z}$ we have $F_{H\mathbb{Z}}(x, y) = x + y$, so the formal group is $\hat{\mathbb{G}}_a$.

For $E = KU$ we have

$$F_{KU}(x, y) = \beta^{-1}\big((\beta x + 1)(\beta y + 1) - 1\big) = x + y + \beta xy.$$

The power series $f(x) = \beta x$ is an isomorphism from F_{KU} to the multiplicative formal group law, since

$$f(x + y + \beta xy) = \beta x + \beta y + (\beta x)(\beta y) = f(x) + f(y) + f(x)f(y),$$

and $f'(0) = \beta$ is a unit. Hence the formal group of KU is $\hat{\mathbb{G}}_m$. Note that this isomorphism is not strict.

As is usually the case when working over general rings, the situation differs between characteristic zero and positive characteristic. In characteristic zero, i.e. when R is a $\mathbb{Q}$-algebra, all our local 1-dimensional "multiplications", i.e. formal group laws, turn out to be isomorphic to the additive one.

This should be thought of as an analogue of the following fact.

Lemma 1.26. *Any real Lie group of dimension* 1 *is locally isomorphic to* $(\mathbb{R}, +)$.

Remark 1.27. In fact there are only two connected 1-dimensional Lie groups up to isomorphism: $(\mathbb{R}, +)$ and $(S^1, \cdot)$.

Fix a formal group law $F \in R[[x, y]]$ and let $\partial_1 F$, $\partial_2 F$ denote the partial derivatives in the first and second variable. Our goal is to find a power series $\log_F(x) = x + (\text{higher order terms})$ such that

$$(*) \qquad \log_F(x +_F y) = \log_F(x) + \log_F(y),$$

i.e. a strict isomorphism $F \to \hat{\mathbb{G}}_a$.

Suppose there is such a power series $\log_F$. Then $\log_F(F(z, x)) = \log_F(z) + \log_F(x)$; differentiating in z and setting $z = 0$, since $F(0, x) = x$ and $\log_F'(0) = 1$, we obtain

$$\log_F'(x)\, \partial_1 F(0, x) = 1.$$

Note that $\partial_1 F(0, x)$ is invertible in $R[[x]]$: its constant term is $\partial_1 F(0, 0) = 1$, because $F(z, 0) = z$. This motivates the following definition.

Definition 1.28. The *invariant differential* of F is

$$\omega_F(x) := \big(\partial_1 F(0,x)\big)^{-1} \in R[[x]].$$

Note that $\omega_F(0) = 1$.

Over a $\mathbb{Q}$-algebra we can integrate, so the candidate $\log_F$ can be constructed as the antiderivative of the invariant differential.

$$\text{if } \omega_F(x) = 1 + \sum_{i\geq 1} a_i x^i, \text{ then}$$
$$\log_F(x) \coloneqq \int_0^x \omega_F = x + \sum_{i\geq 1} \frac{a_i}{i+1} x^{i+1}$$

The only remaining question is whether the antiderivative of ω_F actually satisfies $(*)$. The key input is the following identity, which explains why ω_F deserves to be called invariant.

Lemma 1.29. $\omega_F(F(x,y))\,\partial_1 F(x,y) = \omega_F(x)$.

Proof. Differentiate the associativity identity $F(z,F(x,y)) = F(F(z,x),y)$ with respect to z:

$$\partial_1 F(z,F(x,y)) = \partial_1 F(F(z,x),y) \cdot \partial_1 F(z,x).$$

Setting $z = 0$ and using $F(0,x) = x$ gives

$$\partial_1 F(0,F(x,y)) = \partial_1 F(x,y) \cdot \partial_1 F(0,x).$$

Inverting both sides gives the claim. □

The lemma says that the translation of $\omega_F(x)\,dx$ along $\tau_y \coloneqq F(-,y)$ is $\omega_F(x)\,dx$ again. So $\omega_F\,dx$ is a translation-invariant 1-form.

Theorem 1.30. *Let R be a $\mathbb{Q}$-algebra and F a formal group law over R. Then there is a unique strict isomorphism $\log_F\colon F \to \hat{\mathbb{G}}_a$, namely*

$$\log_F(x) = \int_0^x \omega_F.$$

In particular, any two formal group laws over a $\mathbb{Q}$-algebra are isomorphic.

Proof. By the discussion above any strict isomorphism $\log_F$ has derivative ω_F and vanishes at 0, so it is unique and given by the antiderivative of ω_F.

Let us now check that $\log_F$ defined above satisfies all required properties. Since $\log_F(0) = 0$ and $\log_F'(0) = \omega_F(0) = 1$, $\log_F$ is invertible under composition and strict.

It remains to check $(*)$, consider

$$H(x,y) := \log_F(F(x,y)) - \log_F(x) - \log_F(y).$$

By the chain rule and Lemma 1.29,

$$\partial_x H = \omega_F(F(x,y))\,\partial_1 F(x,y) - \omega_F(x) = 0,$$

and $H(0,y) = \log_F(y) - 0 - \log_F(y) = 0$ since $F(0,y) = y$. It is straightforward to check that if a power series $H(x,y) \in R[[x,y]]$ satisfies $H(0,y) = \partial_x H = 0$ then it must be zero, which precisely shows $(*)$. □

2. Appendix: Standard picture from the theory of Lie groups

First, consider a smooth manifold M. There are several ways to define the latter: the classical one, using charts and atlases, the functor of points perspective (which embeds the previous definition into a certain category of functors), and the sheaf-theoretic perspective.

Considering any of these definitions, we think of a smooth manifold as a topological space M which locally "looks like" $\mathbb{R}^n$ for some $n \in \mathbb{N}$, with these local data glued together in a smooth way.

The classical definition of a tangent bundle TM of a smooth manifold M is given by:

Definition 2.1. The tangent bundle $TM \to M$ is a smooth vector bundle whose fiber over $p \in M$ is the vector space

$$T_pM = \{[\gamma] \mid \gamma : (-\varepsilon, \varepsilon) \to M,\ \gamma(0) = p\},$$

where two curves γ_1 and γ_2 are equivalent if, for some (and hence any) local coordinate system around p, their coordinate representations have the same derivative at 0.

However, there is another, more algebraic, way to define the tangent space. Let $C^\infty(M)$ denote the algebra of smooth real-valued functions on M. For a point $p \in M$, we regard $\mathbb{R}$ as a $C^\infty(M)$-module via evaluation at p, that is,

$$f \cdot a = f(p)a, \qquad f \in C^\infty(M), \quad a \in \mathbb{R}.$$

Definition 2.2. A *derivation at p* is an $\mathbb{R}$-linear map

$$D : C^\infty(M) \longrightarrow \mathbb{R}$$

satisfying the Leibniz rule

$$D(fg) = f(p)D(g) + g(p)D(f)$$

for all $f, g \in C^\infty(M)$.

The tangent space of M at p is the vector space of derivations at p:

$$T_pM = \mathrm{Der}_p(C^\infty(M), \mathbb{R}).$$

The tangent bundle is then defined as the disjoint union

$$TM = \coprod_{p \in M} T_pM,$$

equipped with the unique smooth structure for which the natural projection

$$\pi : TM \longrightarrow M, \qquad D \longmapsto p,$$

is a smooth vector bundle.

The equivalence of these two definitions is expressed by the following theorem.

Theorem 2.3. *Let M be a smooth manifold and $p \in M$. There is a canonical isomorphism*

$$T_pM_{\mathrm{curves}} \cong T_pM_{\mathrm{der}},$$

given by

$$[\gamma] \longmapsto D_\gamma, \qquad D_\gamma(f) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)).$$

These isomorphisms assemble to a canonical isomorphism of smooth vector bundles

$$TM_{\mathrm{curves}} \cong TM_{\mathrm{der}}.$$

Sketch of proof. We will show only an easy direction. Let $\gamma : (-\varepsilon, \varepsilon) \to M$ be a smooth curve with $\gamma(0) = p$. For every $f \in C^\infty(M)$, define

$$D_\gamma(f) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)).$$

By the ordinary product rule,

$$\begin{aligned} D_\gamma(fg) &= \left.\frac{d}{dt}\right|_{t=0} \big(f(\gamma(t))g(\gamma(t))\big) \\ &= f(p)D_\gamma(g) + g(p)D_\gamma(f). \end{aligned}$$

Hence D_γ is a derivation at p. □

Therefore, the previous theorem says that derivations of functions on the manifold are all derivations along 1-dimensional curves. The derivation perspective is particularly useful because it makes the tangent space intrinsic: a tangent vector is an operation which differentiates smooth functions at a point.

Definition 2.4. A *Lie group* is a group object in the category of smooth manifolds.

Multiplication by a fixed element $g \in G$ gives a bijection between the tangent spaces at the identity and at g. Thus, the whole tangent bundle is isomorphic to the product $G \times T_eG$. The tangent space at the identity is denoted by:

$$\mathfrak{g} := T_eG = Der_e(C^\infty(G), \mathbb{R}),$$

and called the *Lie algebra* of G.

In fact, to be eligible for the name "Lie algebra", the tangent space at the identity must be equipped with a Lie bracket; let us fill this gap.

Since all fibers of the tangent bundle are isomorphic via translation, we can produce a vector field on G from any tangent vector at the identity. (For now it suffices to think of a vector field as a smooth assignment of a tangent vector to each point of the manifold.) This is done by taking an arbitrary tangent vector $v \in T_eG$ and translating it to every point.

Therefore any derivation d in $\mathfrak{g} = Der_e(C^\infty(G), \mathbb{R})$, in fact, gives rise to a function: $C^\infty(G) \xrightarrow{\tilde{d}} C^\infty(G)$, by the following procedure:

(1) fix a derivation $d \in Der_e(C^\infty(G), \mathbb{R})$, and thus, the whole set of derivations $d_g \in Der_g(C^\infty(G), \mathbb{R})$.
(2) then, we take a smooth function $f \in C^\infty(G)$
(3) and define a new function $\tilde{d}(f) \in C^\infty(G)$: $\tilde{d}(f)(g) = d_g(f)$

Derivations with the signature $C^\infty(G) \to C^\infty(G)$ are called (or may be identified with) *vector fields* on G. The vector fields coming from the tangent vectors at the identity are precisely the *left-invariant vector fields* on G.

The set of vector fields is naturally equipped with a Lie bracket, (coming from the ring structure on $C^\infty(G)$):

$$[X, Y](f) = X(Y(f)) - Y(X(f)), \qquad f \in C^\infty(G)$$

which is left-invariant if X and Y are left-invariant and can be restricted to the tangent space at the identity, thus giving a Lie bracket on $\mathfrak{g}$.

Definition 2.5. The vector space

$$\mathfrak{g} = T_eG$$

equipped with the bracket

$$[v, w] := ev_e \circ [\widetilde{v}, \widetilde{w}] \in Der_e(C^\infty(G), \mathbb{R}) \cong T_eG$$

is called the *Lie algebra* of the Lie group G, where $\widetilde{v}$ and $\widetilde{w}$ are the left-invariant vector fields corresponding to v and w.

The Jacobi identity for this bracket follows from the Jacobi identity for the commutator of vector fields. Hence

$$(\mathfrak{g}, [-,-])$$

is a Lie algebra.

Let $\dim G = n$ and choose a coordinate chart, $U \subseteq G$:

$$\varphi : U \longrightarrow V \subseteq \mathbb{R}^n$$

such that

$$\varphi(e) = 0.$$

Write

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n)$$

for two points in V. Whenever x and y are sufficiently close to 0, the product of the corresponding points of G is again contained in U, and hence multiplication is represented in these coordinates by a smooth map

$$F : V \times V \longrightarrow \mathbb{R}^n,$$

defined by

$$F(x,y) = \varphi\left(\varphi^{-1}(x)\varphi^{-1}(y)\right).$$

Since e is the identity, the multiplication satisfies

$$F(x,0) = x, \qquad F(0,y) = y.$$

Associativity of the group multiplication becomes

$$F(F(x,y),z) = F(x,F(y,z)).$$

The map F is smooth, and therefore it has a Taylor expansion around $(0,0)$. The conditions

$$F(x,0) = x, \qquad F(0,y) = y$$

imply that its expansion starts with

$$F(x,y) = x + y + \text{terms of degree at least 2.}$$

The expression

$$F(x,y)$$

is called a *formal group law* if one forgets convergence and regards the Taylor series as a formal power series

$$F(X,Y) \in \mathbb{R}[[X_1, \ldots, X_n, Y_1, \ldots, Y_n]]^n.$$

The identities satisfied by the group multiplication become identities of formal power series:

$$F(X,0) = X, \qquad F(0,Y) = Y,$$

and

$$F(F(X,Y),Z) = F(X,F(Y,Z)).$$

Thus the formal group law is the formal Taylor expansion of the local multiplication law of the Lie group at the identity.

As usual in algebraic geometry, the story about formal group laws is obtained by replacing the base field $\mathbb{R}$ by an arbitrary commutative ring R. The more surprising fact is that formal group laws arise in the context of cohomology theories.

Remark 2.6. The specific form of the Taylor series depends on the chosen coordinates. There is a canonical way to choose the coordinates. Using the exponential map $\exp : \mathfrak{g} \to G$ from the Lie algebra $\mathfrak{g} = T_eG$, we obtain a local diffeomorphism. Since the tangent space at the identity is a vector space of the same dimension as the Lie group, this precisely gives a local coordinate system around the identity.

The multiplication is locally governed by:

$$Z \coloneqq F(X,Y) = \exp^{-1}(\exp(X)\exp(Y)) \quad \text{compare this to the general formula}$$

where $X, Y, Z \in \mathfrak{g}$.

The function determining Z from X and Y is given by the famous (and beautiful) Baker–Campbell–Hausdorff (BCH) formula.

Abstractly, the BCH theorem provides a universal series in the graded completion of the free Lie algebra generated by two non-commuting variables X and Y, expressed entirely via iterated commutators:

$$Z = \mathrm{BCH}(X,Y) = X + Y + \frac{1}{2}[X,Y] + \frac{1}{12}([X,[X,Y]] - [Y,[X,Y]]) - \frac{1}{24}[Y,[X,[X,Y]]] + \dots$$

To see the actual formal group law from a BCH formula (i.e. not involving Lie brackets), we need to choose a coordinate basis.

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